

Mohammad Shaharyar Ahsan

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EDUCATION

Lahore University Of Management Sciences (LUMS)

Lahore, Pakistan

B.S. Computer Science, Current GPA: 3.80, Major GPA: 3.87

Sept. 2021 – June 2025

– **Awards:** Dean's Honor List (2022–2025) & Award of High Distinction

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, Kotlin, Bash, HTML/CSS

Frameworks & ML: React.js, Node.js, Next.js, FastAPI, LangChain, Amazon Bedrock, Pinecone, Pandas, Scikit-learn, Unsloth

Tools & Cloud: AWS (KMS), Vercel, DigitalOcean, Docker, Git, REST APIs, OAuth, BLE GATT Protocols, Firmware

WORK EXPERIENCE

Developer Advocate

June 2025 – Present

Educative

Lahore, Pakistan

- Engineered a high-volume executive content pipeline and organic distribution engine, driving 3.4M+ combined LinkedIn impressions, 33K+ clicks, and acquiring 7,400+ new followers with improved 5.9% CTRs.
- Developed custom Python data scrapers and a proprietary Chrome Extension to automate internal enterprise workflows, eliminating manual data entry hurdles and saving 20+ operational hours monthly.
- Orchestrated a data-driven reputation turnaround via customer sentiment analysis, elevating Trustpilot scores from 2.5 to 4.1.
- Executed technical lifecycle marketing and '30 Days of Code' giveaway campaigns, converting ~7,000 new funnel participants, increasing site sessions by 595%, and generating 1.5M+ email opens (¿25% open rate).

Research Assistant

June 2024 – July 2025

Embedded AI Lab, LUMS

Lahore, Pakistan

- Architected a low-cost IoT health monitoring ecosystem utilizing an ATmega328P MCU; engineered C/C++ firmware with optimal duty-cycled sampling to maximize battery longevity during 96-hour clinical trials.
- Developed a multi-threaded Java Android application leveraging BLE GATT protocols for secure background data bridging between the wearable hardware and a distributed Node.js inference server.
- Engineered an LLM sequence-to-values interpreter and Pinecone RAG pipeline with custom tool-calling (cron scheduling, clinical chart retrieval), reducing Heart Rate MAE by 46% while ensuring 100% data availability.
- Implemented a dynamic LLM routing system utilizing GPT-4o-mini and o1-mini to amortize API costs based on user query complexity, achieving a 56.5% relative cost reduction.

Research Intern, NLP

Jan 2024 – May 2024

Center for Speech & Language Technologies, LUMS

Lahore, Pakistan

- Engineered a Python scraping pipeline utilizing the Reddit API to construct a conversational, structurally-aligned Alpaca dataset from the r/AskDocs subreddit using Regex preprocessing.
- Fine-tuned a Gemma foundational LLM utilizing the Unsloth framework and LoRA adapters on a pooled dataset, enabling scalable and localized healthcare inference capabilities for resource-constrained environments.

PROJECTS

Hooked on a Feeling: Computational Linguistics

Sept 2024 – Dec 2024

Publication Link

LUMS

- Built a multi-stage distributed scraping pipeline (Python, BeautifulSoup, Dask) to aggregate a musicological dataset spanning 353 Grammy nominees across seven decades.
- Engineered NLP pipelines using fine-tuned DistilBERT inference and TF-IDF thematic modeling to isolate sentiment shifts, identifying a statistically significant macroeconomic trend of 'multimodal dissonance'.

PUBLICATIONS

Hooked On A Feeling: A Computational Approach Towards Understanding Songwriting As Cultural Memory –

Published in Digital Studies (DSCN) [DOI: 10.16995/dscn.23539]

LLM-Enhanced Wearables for Comprehensible Health Guidance in LMICs – Pre-print [arXiv: 2602.08701]

Poster: Bridging IoT Gaps in Developing Regions with LLMs – EWSN '24 [PDF]